FINAL PROJECT REPORT

SUBMITTED BY

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COLLEGE NAME

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BRANCH

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

ROLL ID

B.TECH 41

PROJECT NAME

Global AI Adoption in Education: A Comprehensive Global Performance Intelligence Report

Final Project Report Template

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    1.2. Objectives
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Final Project Report

1. Introduction

1.1 Project Overview

The **Global AI Adoption in Education** project analyzes the adoption and usage of Artificial Intelligence in the education sector. The project uses Tableau to convert the dataset into interactive visualizations and dashboards.

The analysis focuses on AI adoption across **countries and regions**, differences between **students and teachers**, **rural and urban AI usage**, **school AI adoption**, **leading AI tools**, **average daily AI usage**, and **year-wise adoption trends**.

1.2 Objectives

- To analyze global AI adoption in education.
- To compare AI usage between students and teachers.
- To identify countries and regions with higher and lower AI adoption.
- To compare rural and urban AI usage.
- To identify leading AI tools used in education.
- To analyze AI adoption trends over time.
- To develop an interactive Tableau dashboard.
- To present meaningful insights through Tableau Story.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	Education policymaker	Understand AI adoption across regions	It is difficult to compare adoption levels	AI adoption varies between countries and regions	Uncertain about where support is needed
PS-2	School administrator	Understand AI adoption in education	Student, teacher, and school usage can differ	AI adoption varies across educational environments	Unsure whether AI adoption is effective
PS-3	Education decision-maker	Identify rural and urban AI adoption differences	Adoption may vary by location	Technology access differs between areas	Concerned about the digital adoption gap

2.2 Project Proposal – Proposed Solution

Section	Details
Objective	Analyze global AI adoption in education using Tableau and identify important adoption patterns and trends.
Scope	Country-wise, region-wise, student, teacher, school, rural/urban, AI tool, daily usage, and year-wise analysis.
Problem Statement	Education stakeholders need an easy way to understand and compare AI adoption across countries, regions, schools, students, and teachers.

Impact	The analysis can help identify adoption gaps and provide insights for education planning and AI implementation.
Section	**Details**
Approach	Collect, clean, prepare, analyze, and visualize the dataset using Tableau.
Key Features	KPI cards, bar charts, line charts, geographical map, filters, comparisons, interactive dashboard, and Tableau Story.

2.3 Initial Project Planning

Sprint	Epic	User Story	Story Points	Priority	Team	Start Date	End Date
Sprint-1	Data Collection & Cleaning	USN-1 – Collect, understand and clean the AI education dataset.	3	High	Self	17Aug2026	18Aug2026
Sprint-1	Tableau Data Preparation	USN-2 – Prepare and connect the dataset to Tableau.	2	High	Self	18Aug2026	19Aug2026
Sprint-2	AI Adoption Analysis	USN-3 – Analyze AI adoption across countries and regions.	3	High	Self	19Aug2026	20Aug2026
Sprint-2	Student & Teacher Analysis	USN-4 – Compare student and teacher AI usage.	3	High	Self	19Aug2026	20Aug2026

Sprint-2	Dashboard Development	USN-5 – Develop an interactive Tableau dashboard.	5	High	Self	20Aug2026	21Aug2026
Sprint-2	Testing & Reporting	USN-6 – Test the dashboard and prepare insights.	3	High	Self	21Aug2026	22Aug2026

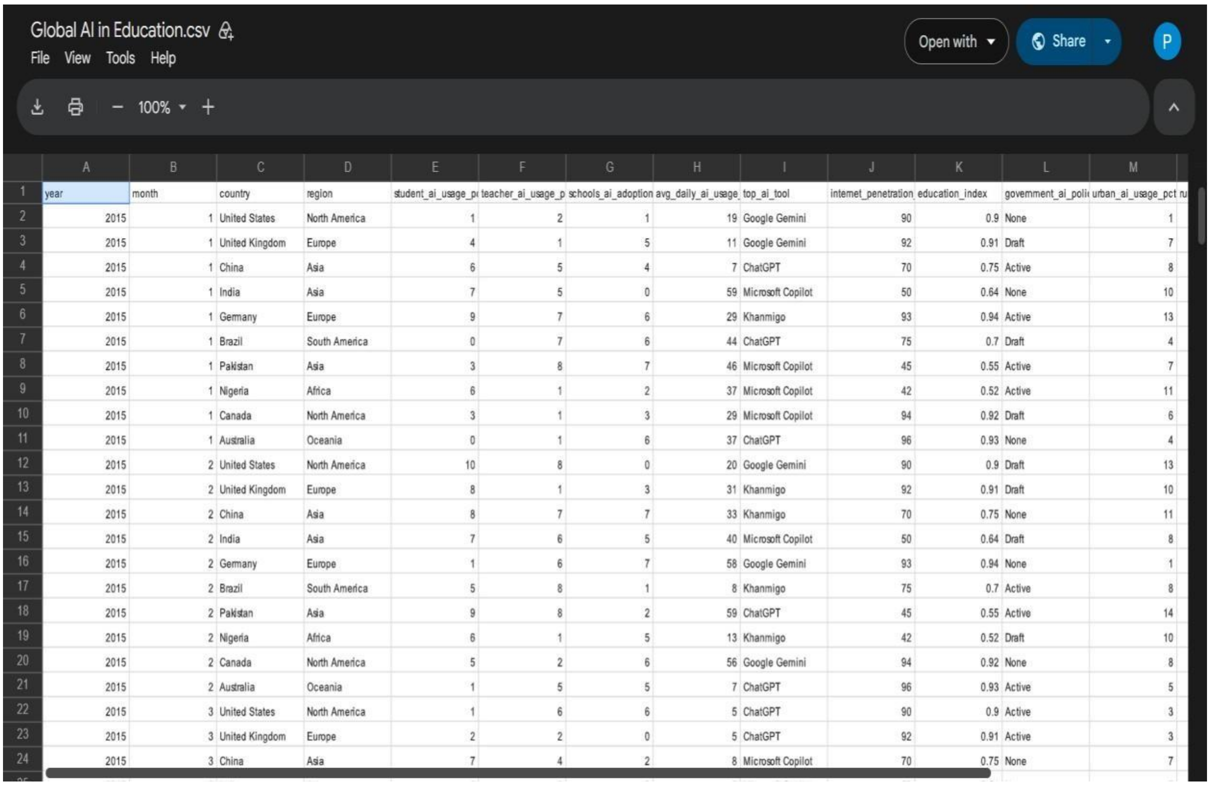
3. Data Collection and Pre-processing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Section	Details
Project Overview	Analyze global AI adoption and usage patterns in education.
Data Collection Plan	Use the project-provided Global AI Adoption in Education dataset and prepare it for Tableau.
Raw Data Source	Global AI Adoption in Education Dataset
Format	CSV
Access	Project-provided / Public dataset

Raw Data Source Screenshot

Raw Dataset



	A	B	C	D	E	F	G	H	I	J	K	L	M
1	year	month	country	region	student_ai_usage_p	teacher_ai_usage_p	schools_ai_adoption	avg_daily_ai_usage	top_ai_tool	internet_penetration	education_index	government_ai_poli	urban_ai_usage_pct
2	2015	1	United States	North America	1	2	1	19	Google Gemini	90	0.9	None	1
3	2015	1	United Kingdom	Europe	4	1	5	11	Google Gemini	92	0.91	Draft	7
4	2015	1	China	Asia	6	5	4	7	ChatGPT	70	0.75	Active	8
5	2015	1	India	Asia	7	5	0	59	Microsoft Copilot	50	0.64	None	10
6	2015	1	Germany	Europe	9	7	6	29	Khanmigo	93	0.94	Active	13
7	2015	1	Brazil	South America	0	7	6	44	ChatGPT	75	0.7	Draft	4
8	2015	1	Pakistan	Asia	3	8	7	46	Microsoft Copilot	45	0.55	Active	7
9	2015	1	Nigeria	Africa	6	1	2	37	Microsoft Copilot	42	0.52	Active	11
10	2015	1	Canada	North America	3	1	3	29	Microsoft Copilot	94	0.92	Draft	6
11	2015	1	Australia	Oceania	0	1	6	37	ChatGPT	96	0.93	None	4
12	2015	2	United States	North America	10	8	0	20	Google Gemini	90	0.9	Draft	13
13	2015	2	United Kingdom	Europe	8	1	3	31	Khanmigo	92	0.91	Draft	10
14	2015	2	China	Asia	8	7	7	33	Khanmigo	70	0.75	None	11
15	2015	2	India	Asia	7	6	5	40	Microsoft Copilot	50	0.64	Draft	8
16	2015	2	Germany	Europe	1	6	7	58	Google Gemini	93	0.94	None	1
17	2015	2	Brazil	South America	5	8	1	8	Khanmigo	75	0.7	Active	8
18	2015	2	Pakistan	Asia	9	8	2	59	ChatGPT	45	0.55	Active	14
19	2015	2	Nigeria	Africa	6	1	5	13	Khanmigo	42	0.52	Draft	10
20	2015	2	Canada	North America	5	2	6	56	Google Gemini	94	0.92	None	8
21	2015	2	Australia	Oceania	1	5	5	7	ChatGPT	96	0.93	Active	5
22	2015	3	United States	North America	1	6	6	5	ChatGPT	90	0.9	Active	3
23	2015	3	United Kingdom	Europe	2	2	0	5	ChatGPT	92	0.91	Active	3
24	2015	3	China	Asia	7	4	2	8	Microsoft Copilot	70	0.75	None	7

3.2 Data Quality Report

Data Source	Data Quality Issue	Severity	Resolution Plan
Global AI Adoption Dataset	Missing or blank values	Moderate	Identify and appropriately handle missing values.
Global AI Adoption Dataset	Duplicate records	Moderate	Check and remove duplicate records.
Global AI Adoption Dataset	Inconsistent category names	Moderate	Standardize text and category values.

Global AI Adoption Dataset	Incorrect data types	Low	Convert fields to appropriate Tableau data types.
Global AI Adoption Dataset	Unnecessary fields	Low	Remove or exclude fields not required for analysis.
Global AI Adoption Dataset	Unusual values	Moderate	Check and validate unusual values before analysis.

Raw Data Source

Attribute	**Description**
Dataset Name	Global AI in Education
File Format	CSV
Number of Records	**1,360**
Number of Attributes	**16**
Time Period	2015–2026
Analysis Tool	Tableau

Dataset Attributes

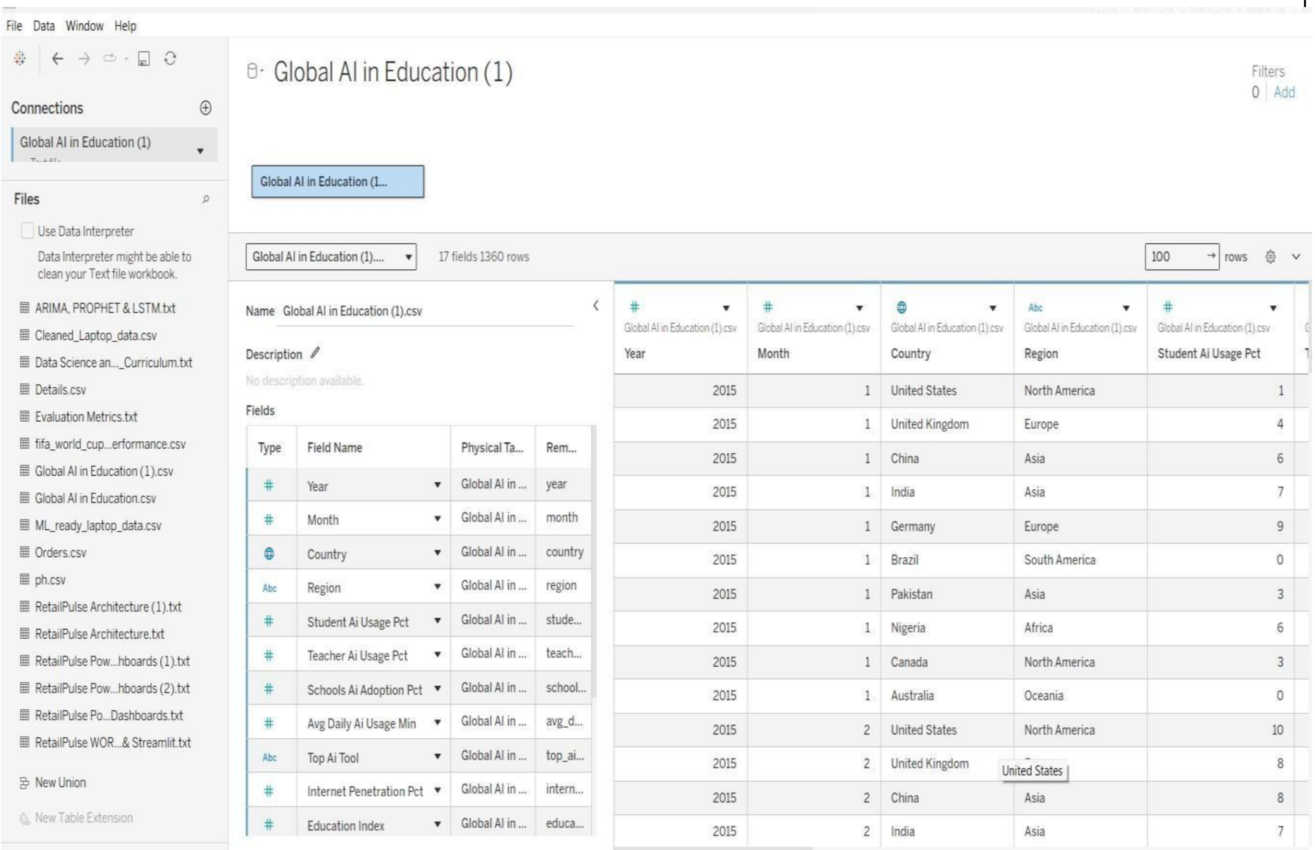
The dataset includes the following fields:

- Year
- Month
- Country
- Region
- Student AI Usage (%)
- Teacher AI Usage (%)
- Schools AI Adoption (%)

- Average Daily AI Usage (Minutes)
- Top AI Tool
- Internet Penetration (%)
- Education Index
- Government AI Policy
- Urban AI Usage (%)
- Rural AI Usage (%)
- AI in Curriculum
- Gender Gap in AI Usage (%)

Data Quality Screenshot

Data Cleaning / Data Quality

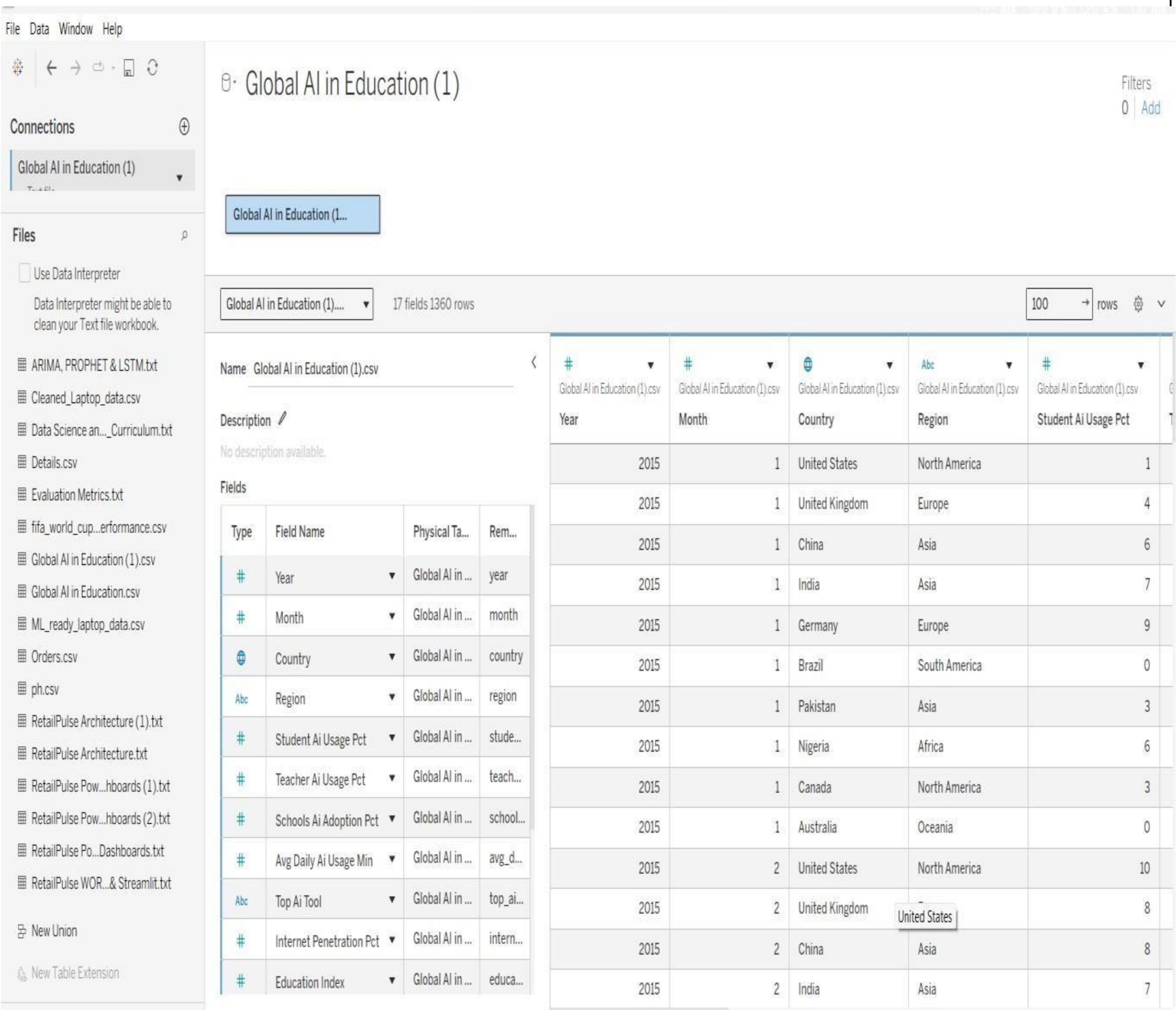


3.3 Data Exploration and Pre-processing

Section	Description
Data Overview	Dataset contains information about AI adoption in education, including student, teacher, school, regional, country, AI tool, rural/urban, and usage indicators.

Data Cleaning	Missing values, duplicates, inconsistent values, and unnecessary fields were checked.
Data Transformation	Data was filtered, sorted, and prepared for Tableau analysis.
Data Type Conversion	Numerical, text, percentage, and date fields were assigned appropriate data types.
Column Splitting/Merging	Fields were reviewed and organized where required.
Data Modeling	Dimensions and measures were prepared for Tableau visualizations and calculated fields.
Save Processed Data	The processed data was connected to and saved within the Tableau workbook.

Pre-processing Screenshot



4. Data Visualization

4.1 Framing Business Questions

The following business questions were developed from the Tableau workbook:

1. What is the overall level of AI adoption in education?
2. How does AI usage differ between students and teachers across countries?

3. Which countries have the highest average daily AI usage?
4. Which AI tools are the leading tools used in education?
5. How has AI usage changed in rural and urban areas over the years?
6. Which regions have the highest school AI adoption?
7. How have school, student, and teacher AI adoption changed over the years?
8. Which countries have higher or lower AI adoption?

4.2 Developing Visualizations

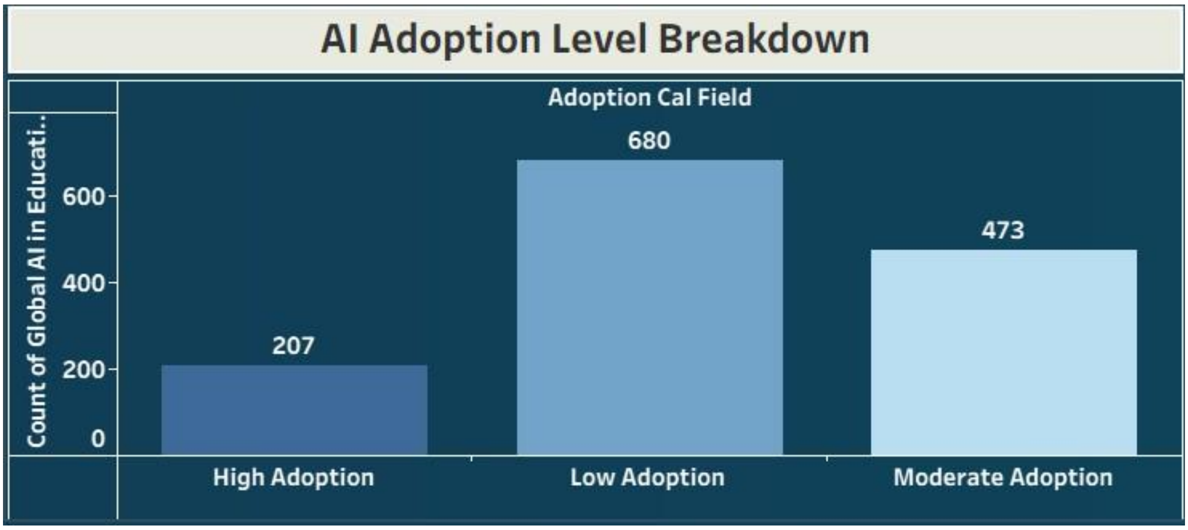
Visualization 1

Business Question: What is the overall level of AI adoption in education?

Visualization: AI Adoption Level Breakdown

Attributes: AI Adoption Level, AI Adoption Percentage

Screenshot:


AI Adoption Level Breakdown
Adoption Cal Field
Count of Global AI in Educati..
600
400
200
0
207
680
473
High Adoption
Low Adoption
Moderate Adoption

Visualization 2

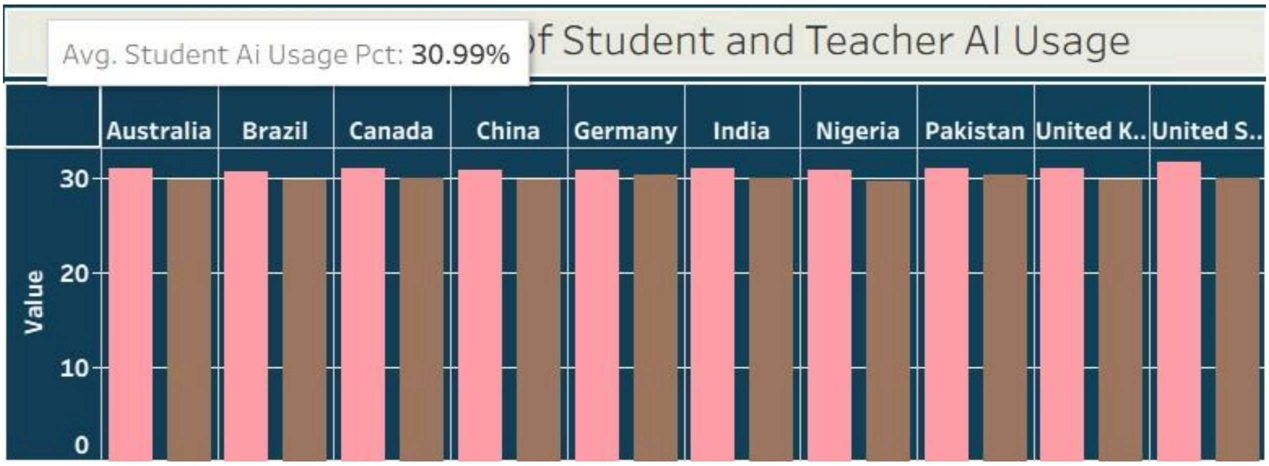
Business Question:

How does AI usage differ between students and teachers across countries?

Visualization: Global Comparision of Student and Teacher AI Usage

Attributes: Country, Student AI Usage, Teacher AI Usage

Screenshot:



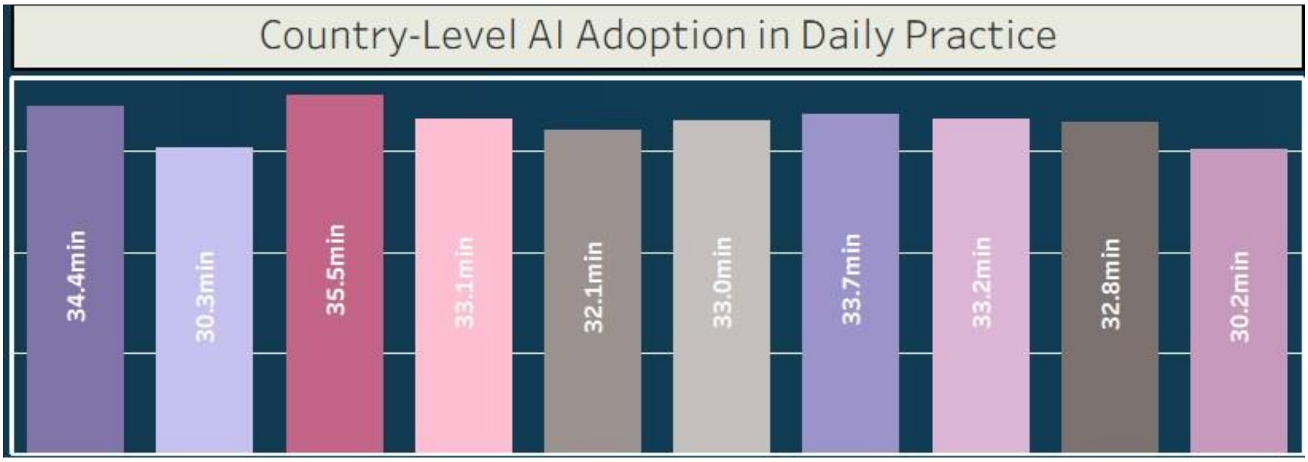
Visualization 3

Business Question: Which countries have the highest average daily AI usage?

Visualization: Country-Level AI Adoption in Daily Practice

Attributes: Country, Average Daily AI Usage

Screenshot:



Which AI tools are the leading tools used in education?

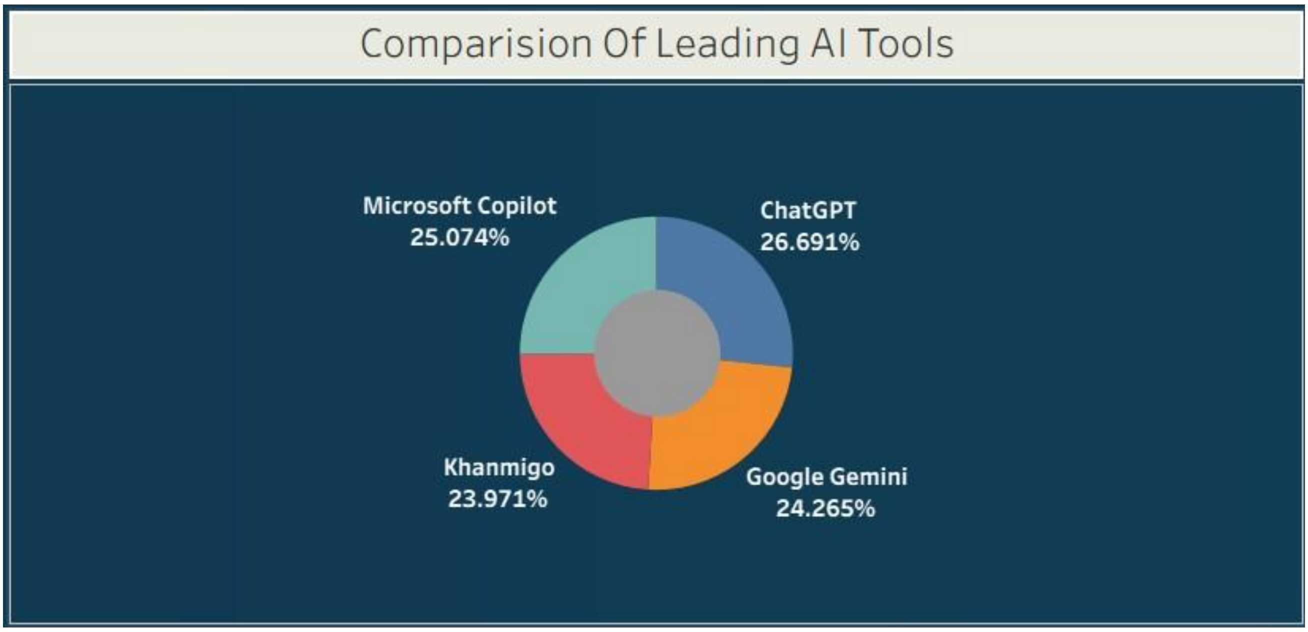
Visualization 4

Business Question:

Visualization: Comparison of Leading AI Tool

Attributes: AI Tool, Tool Usage / Adoption

Screenshot:



Visualization 5

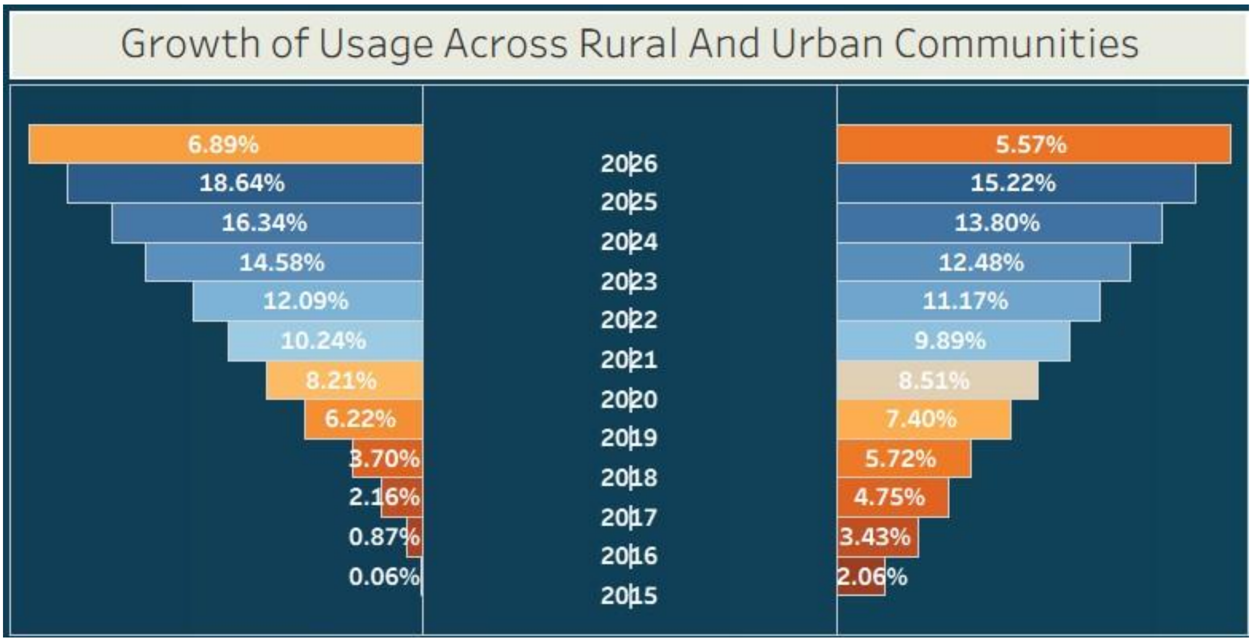
Business Question:

How has AI usage changed in rural and urban areas over the years?

Visualization: Growth of Usage Across Rural And Urban Communities

Attributes: Year, Rural/Urban, AI Usage

Screenshot:



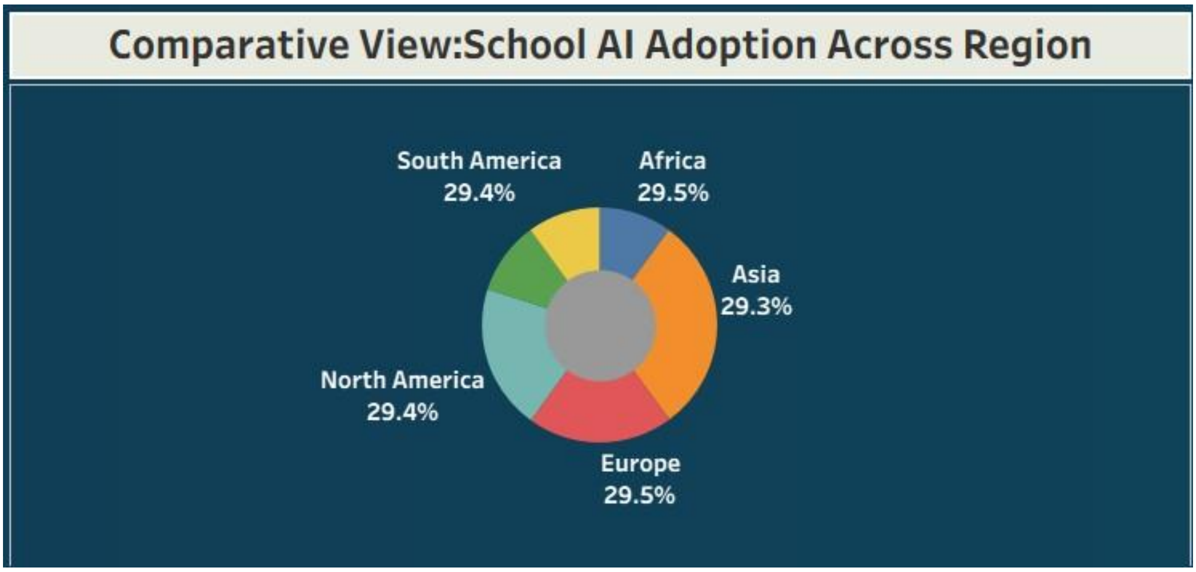
Visualization 6

Business Question:

Which regions have the highest school AI adoption?

Visualization: Comparative View : School AI Adoption Across Region

Attributes: Region, School AI Adoption **Screenshot:**



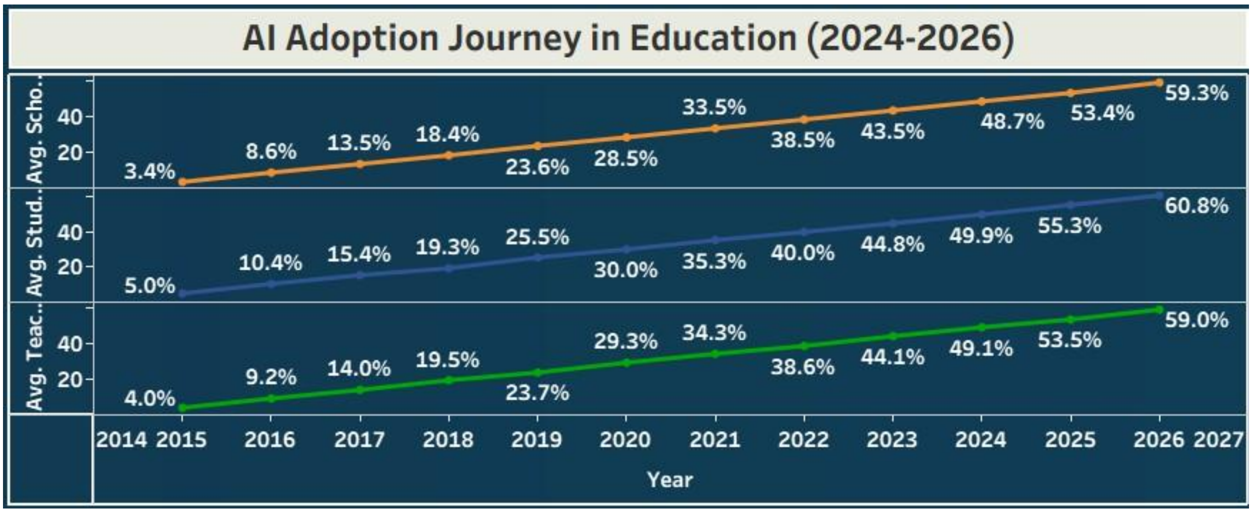
How have school, student, and teacher AI adoption changed over the years?

Visualization 7

Business Question:

Visualization: AI Adoption Journey in Education (2024-2026)

Attributes: Year, School Adoption, Student Adoption, Teacher Adoption **Screenshot:**



Which countries have higher or lower AI adoption?

Visualization: Geographic Trends in Educational Development

Attributes: Country, AI Adoption

Screenshot:

Visualization 8

Business Question:

5. Dashboard

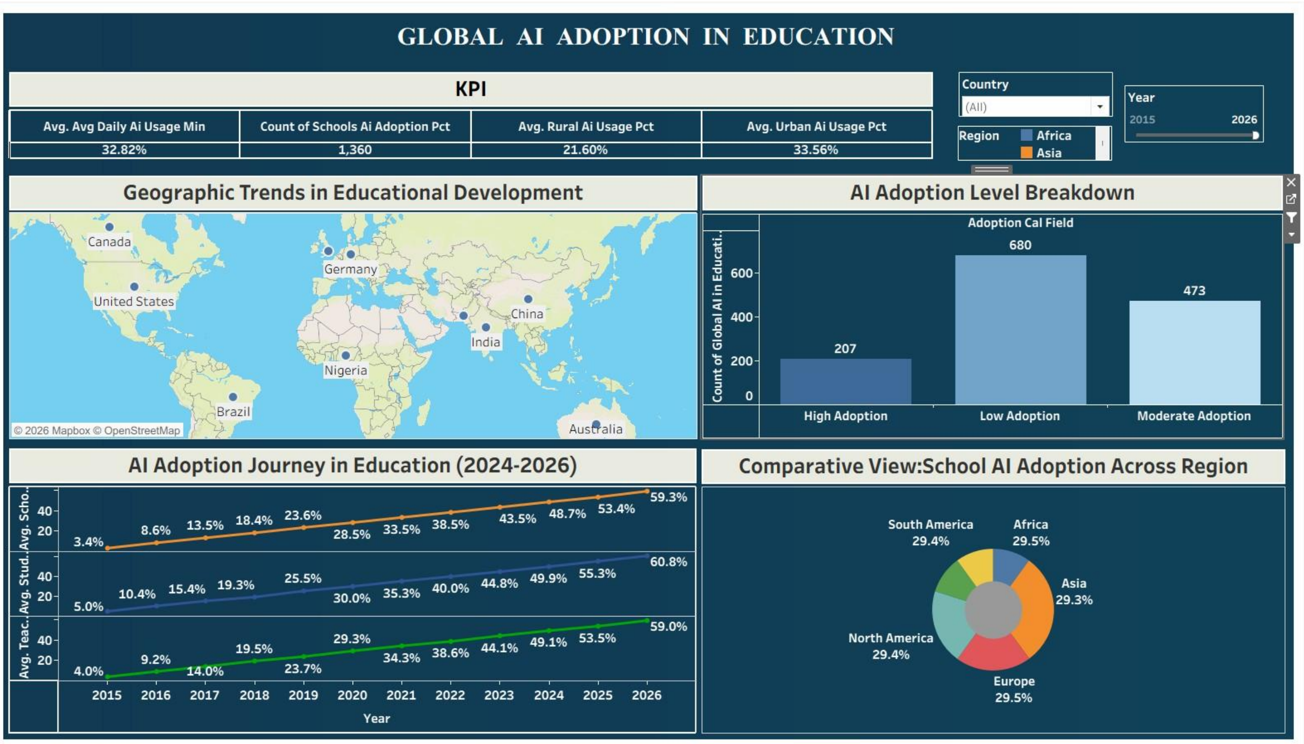
5.1 Dashboard Design File

The dashboard combines KPIs, charts, maps, filters, and comparison visualizations to provide an interactive overview of global AI adoption in education.

Dashboard Design Principles

- Clear and intuitive layout
- Appropriate visualization selection
- Consistent colour and theme
- Interactive filters
- Geographical visualization
- KPI cards
- Comparison charts
- Trend analysis
- Interactive dashboard navigation

- **Dashboard Screenshot**



Additional Dashboard Screenshot

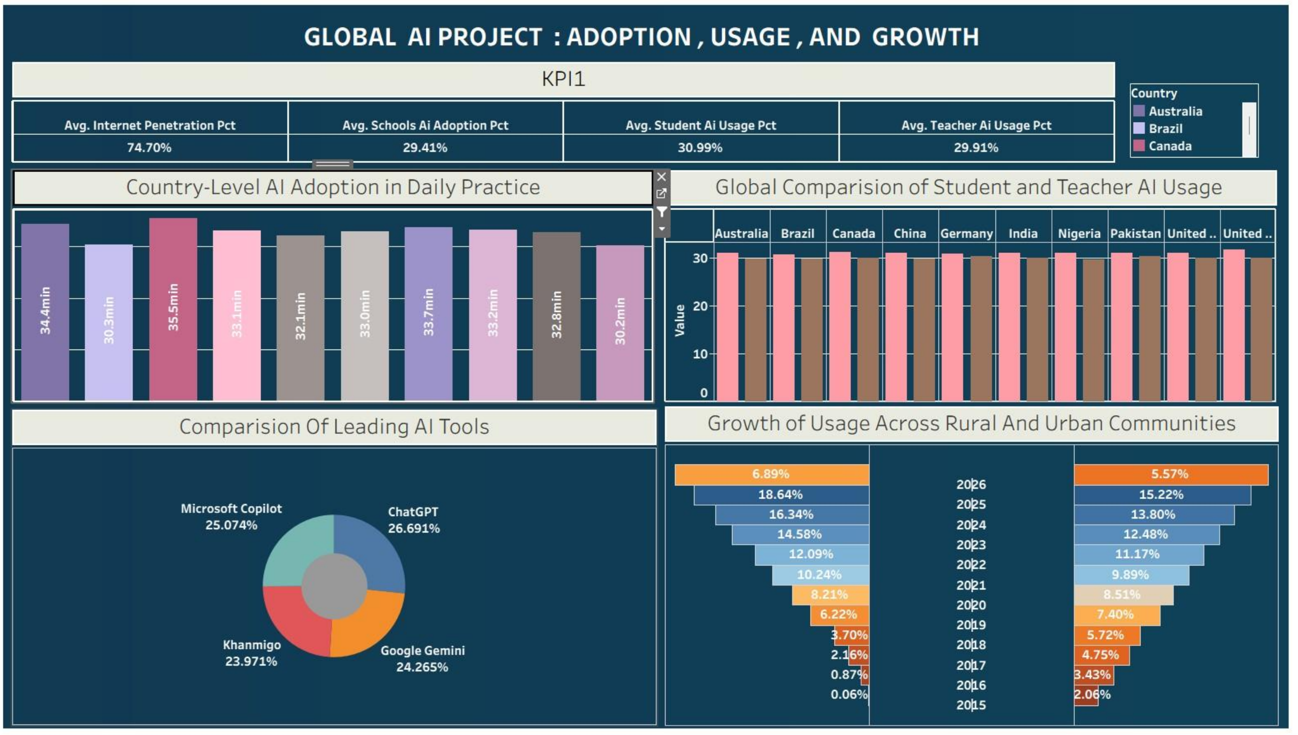


Tableau Public Dashboard Link

Link:

https://public.tableau.com/shared/S7239433G?:display_count=n&:origin=viz_share_link

6. Report

6.1 Story Design File

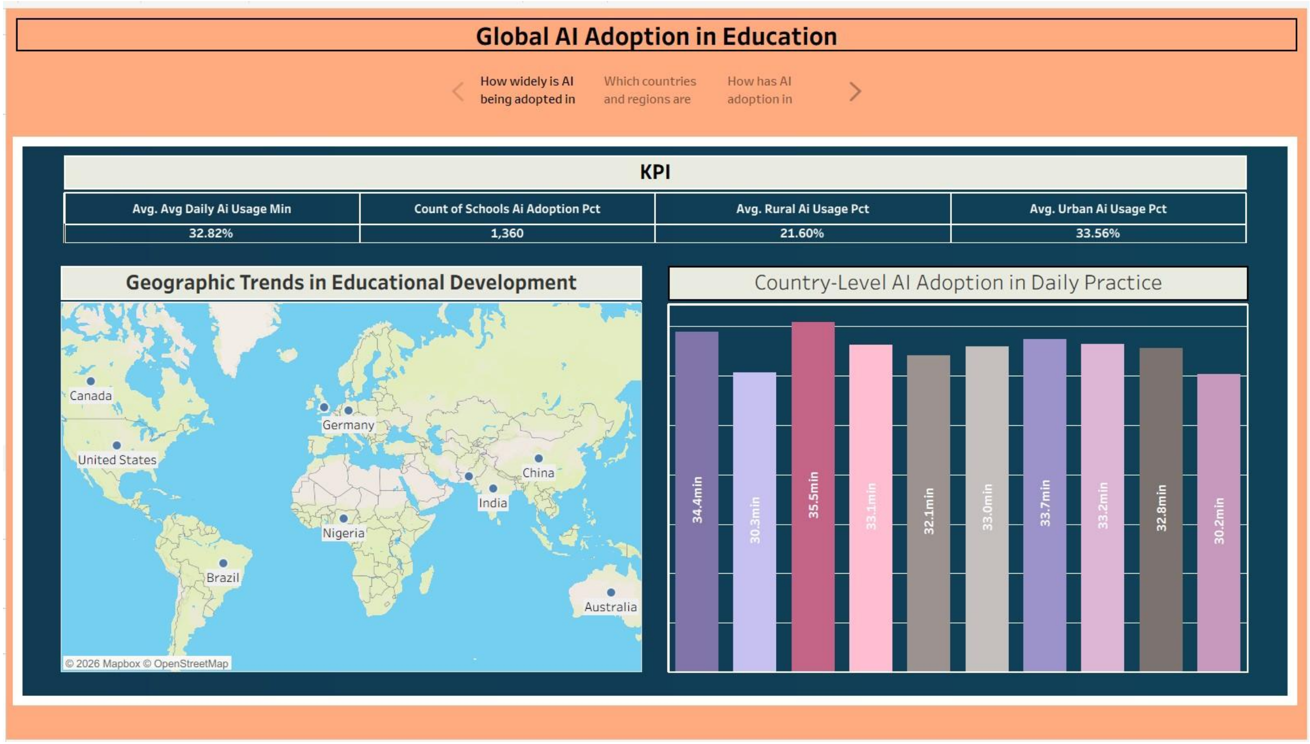
The Tableau Story presents the findings in a logical sequence, beginning with overall adoption and progressing to country, regional, stakeholder, tool, and time-based analysis.

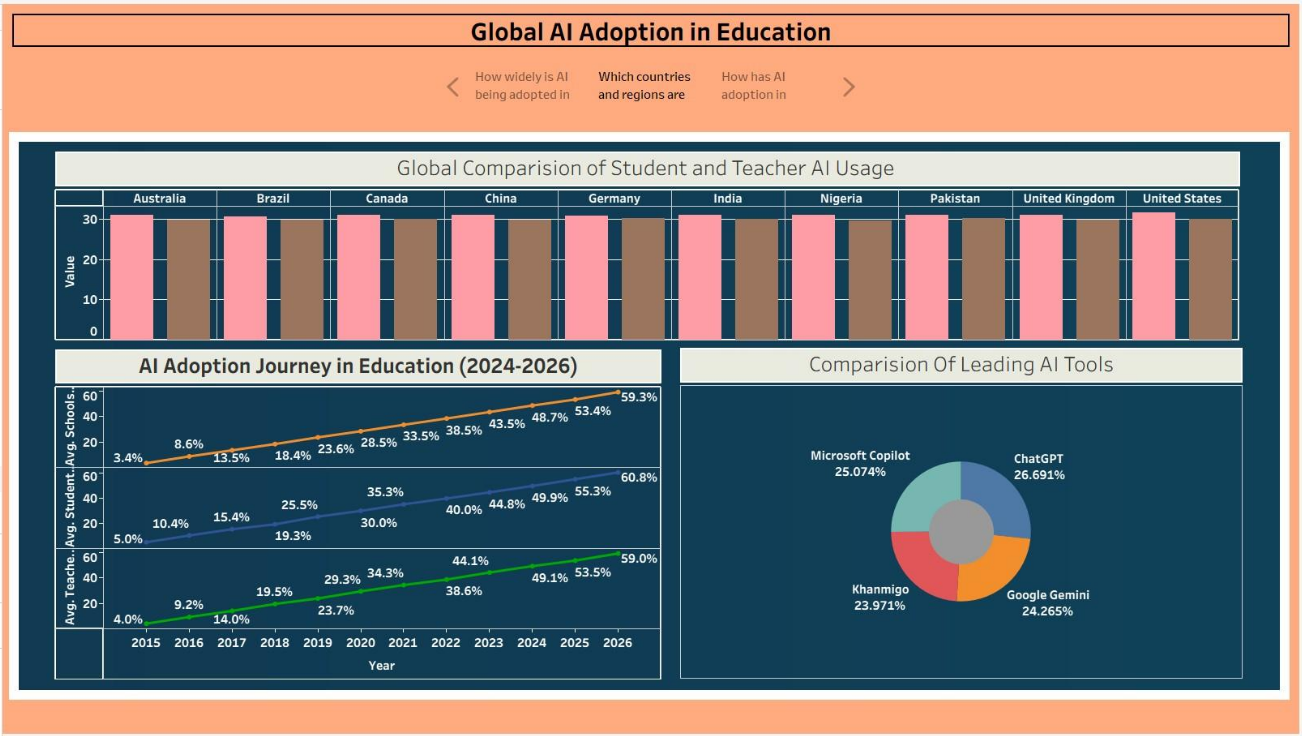
Story Flow

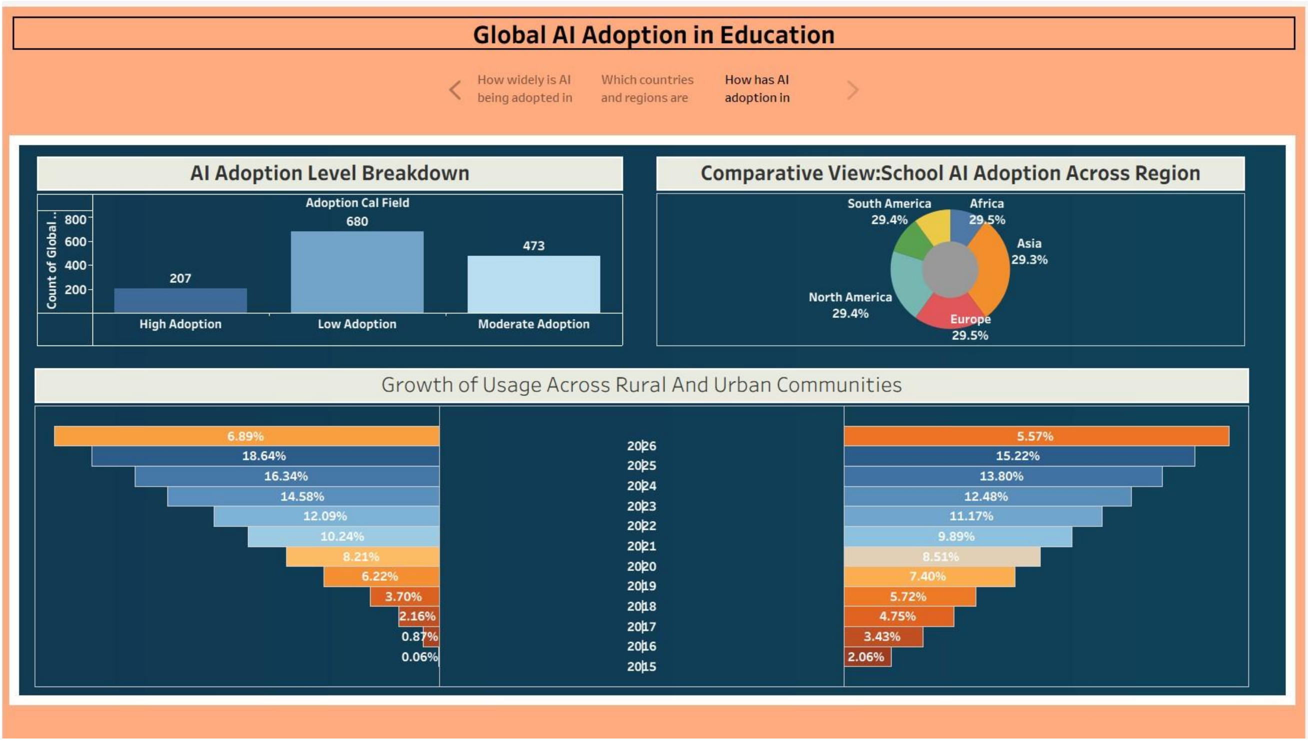
Story Point 1 → Overall AI Adoption
↓
Story Point 2 → Global AI Adoption Map
↓
Story Point 3 → Student & Teacher Usage
↓
Story Point 4 → Average Daily AI Usage
↓
Story Point 5 → Leading AI Tools
↓
Story Point 6 → Rural vs. Urban Growth
↓
Story Point 7 → Regional School Adoption
↓
Story Point 8 → Year-wise School, Student & Teacher Adoption

Story Screenshot

Tableau Story Screenshot


Global AI Adoption in Education
How widely is AI being adopted in
Which countries and regions are
How has AI adoption in
KPI
Avg. Avg Daily Ai Usage Min
32.82%
Count of Schools Ai Adoption Pct
1,360
Avg. Rural Ai Usage Pct
21.60%
Avg. Urban Ai Usage Pct
33.56%
Geographic Trends in Educational Development
Canada
United States
Germany
China
India
Nigeria
Brazil
Australia
© 2026 Mapbox © OpenStreetMap
Country-Level AI Adoption in Daily Practice
34.4min
30.3min
35.5min
33.1min
32.1min
33.0min
33.7min
33.2min
32.8min
30.2min


Global AI Adoption in Education
How widely is AI being adopted in
Which countries and regions are
How has AI adoption in
Global Comparision of Student and Teacher AI Usage
Australia
Brazil
Canada
China
Germany
India
Nigeria
Pakistan
United Kingdom
United States
Value
30
20
10
0
AI Adoption Journey in Education (2024-2026)
Avg. Schools..
Avg. Student..
Avg. Teache..
3.4%
8.6%
13.5%
18.4%
23.6%
28.5%
33.5%
38.5%
43.5%
48.7%
53.4%
59.3%
5.0%
10.4%
15.4%
19.3%
25.5%
30.0%
35.3%
40.0%
44.8%
49.9%
55.3%
60.8%
4.0%
9.2%
14.0%
19.5%
23.7%
29.3%
34.3%
38.6%
44.1%
49.1%
53.5%
59.0%
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
2025
2026
Year
Comparision Of Leading AI Tools
Microsoft Copilot
25.074%
ChatGPT
26.691%
Khanmigo
23.971%
Google Gemini
24.265%



Major Story Observations

- AI adoption differs across countries and regions.
- Student and teacher AI usage can show different adoption patterns.
- Average daily AI usage varies across countries.
- Leading AI tools can be compared to understand tool adoption.
- Rural and urban AI usage shows differences in adoption.
- School AI adoption varies by region.
- School, student, and teacher adoption can be tracked over time.
- The global map provides an easy geographical comparison of AI adoption.

Tableau Public Story Link

Link:

https://public.tableau.com/shared/ZMQRKX4ZQ?:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data Filters

The Tableau dashboard uses interactive filtering to allow users to explore the data according to relevant dimensions.

Filter / Interactive Feature	Purpose
Country	Analyze individual countries
Region	Compare geographical regions
Year	Analyze trends over time
Student / Teacher	Compare stakeholder usage
Rural / Urban	Compare location-based adoption
AI Tool	Analyze leading AI tools

7.2 Number of Calculation Fields

Calculated fields were used where required to support **AI adoption percentages, averages, comparisons, KPIs, and analytical measures**.

Screenshot – Calculated Fields

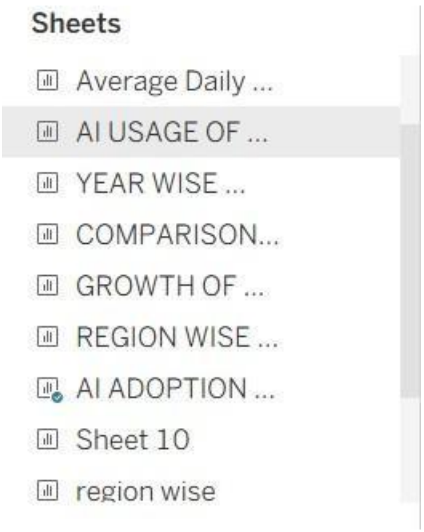
If you open **Data Pane → Calculated Fields** in Tableau, enter the exact number shown there.

7.3 Number of Visualizations

The project contains multiple visualizations covering:

- AI adoption levels
- Student and teacher usage
- Average daily AI usage
- Leading AI tools
- Rural and urban trends
- Regional school adoption
- Year-wise adoption
- Global country-wise adoption **Number of Visualizations: 10**

Screenshot – Tableau Worksheets



8. Conclusion / Observation

The **Global AI Adoption in Education** Tableau project provides an interactive analysis of AI usage across the education sector. The dashboard makes it easier to compare countries, regions, students, teachers, schools, rural and urban areas, and AI tools.

The visualizations demonstrate that AI adoption is not uniform across all educational environments. Country-wise and region-wise differences can be explored through maps and charts, while year-wise analysis helps understand adoption trends.

The project demonstrates how Tableau can transform raw education data into meaningful visual insights that support data-driven decision-making.

Key Outcomes

- Global AI adoption can be explored interactively.
- Student and teacher AI usage can be compared.
- Regional and country-wise adoption differences can be identified.
- Rural and urban adoption can be compared.
- Leading AI tools can be analyzed.
- AI adoption trends can be studied over time.
- Interactive Tableau dashboards provide decision-support insights.

9. Future Scope

- Add more recent AI adoption data.
- Include additional countries and regions.
- Add more education-related indicators.
- Introduce predictive AI adoption analysis.
- Add advanced Tableau calculations and parameters.
- Develop real-time or regularly updated dashboards.
- Add more detailed school-level analysis.
- Compare AI adoption with internet and technology accessibility.
- Publish updated versions of the dashboard on Tableau Public.

10.GitHub & Project Demo Link

GitHub Repository : https://github.com/pranavpatil2092/Nasscom-Project

Dashboard:

Link:

https://public.tableau.com/shared/DHBCK768P?:display_count=n&:origin=viz_share_link

Tableau Public Story

Link:

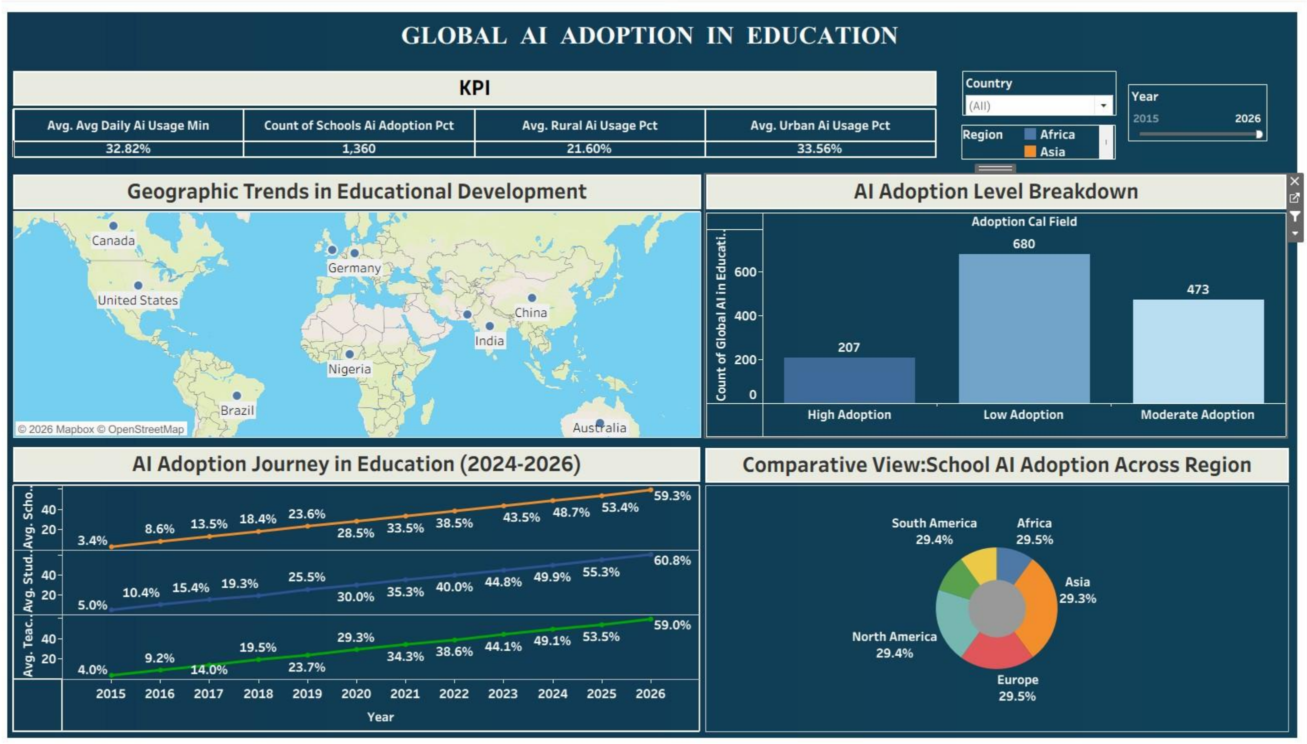
https://public.tableau.com/shared/2T63C3B8W?:display_count=n&:origin=viz_share_link

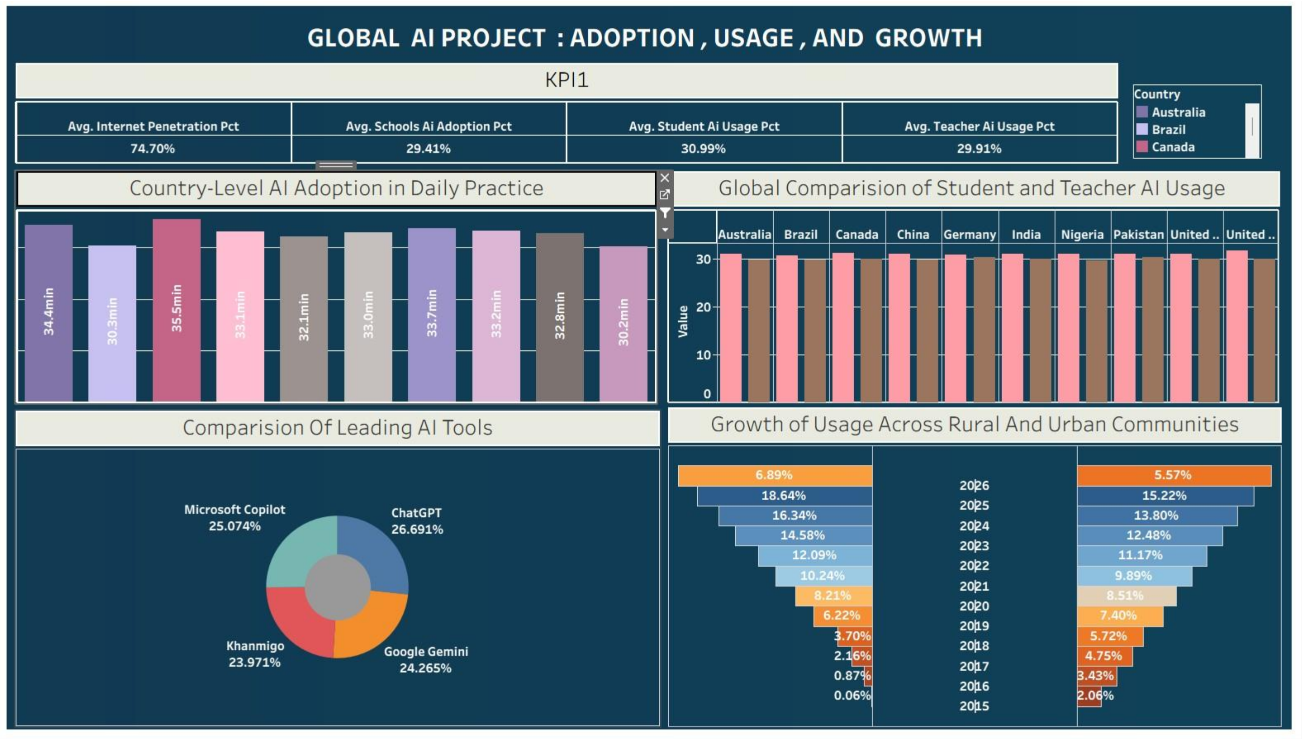
Project Demo Video:

Link:
https://drive.google.com/file/d/1oKEmXXNP3yVZDFx5oMAWhFxvIKigWpy3/view?usp=drive_link

Final Dashboard Screenshot:

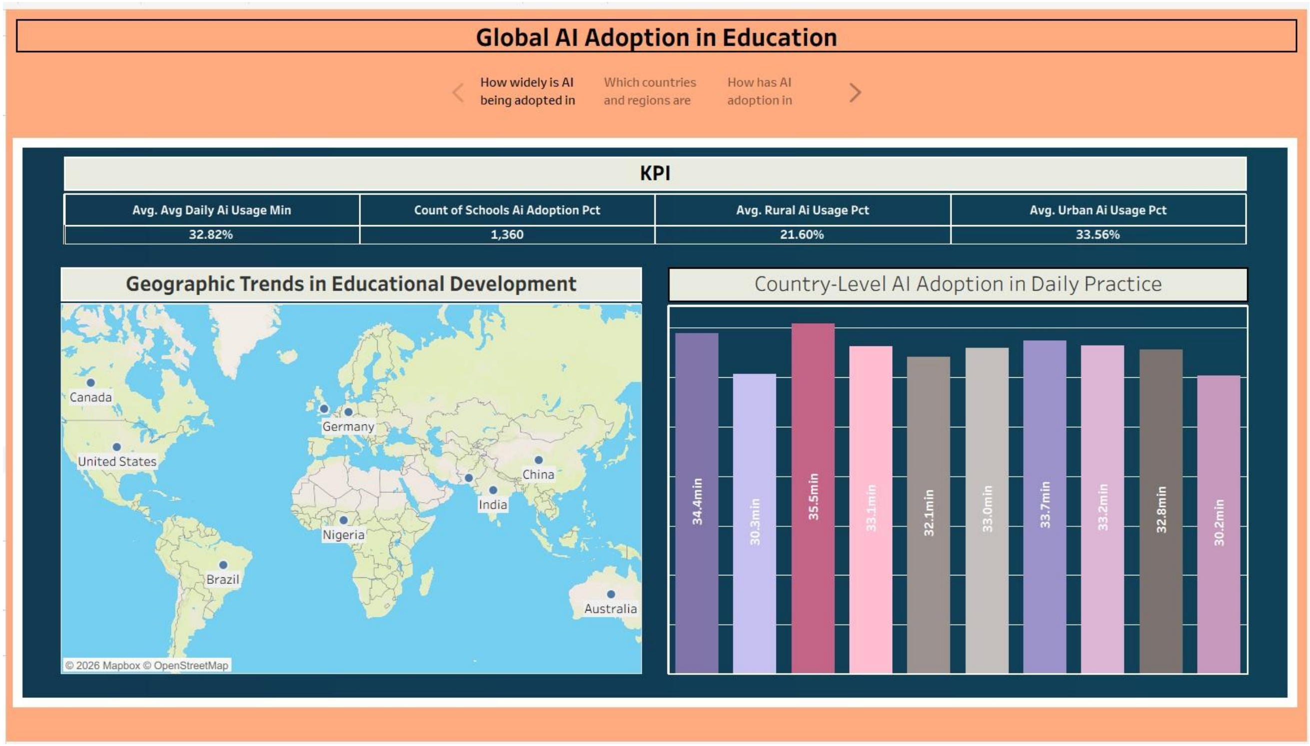
1.



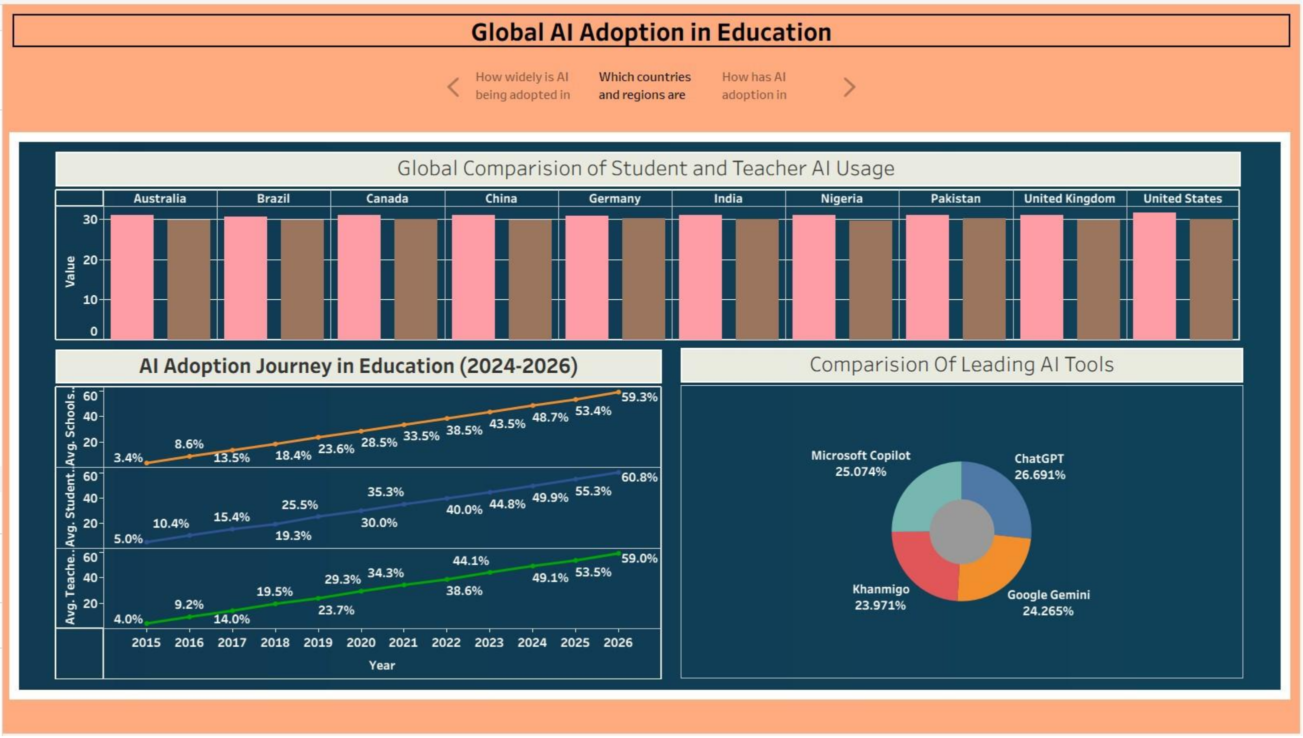


Final Story Screenshot :

1.



2.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How has AI adoption in
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United Kingdom
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Value
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Comparision Of Leading AI Tools
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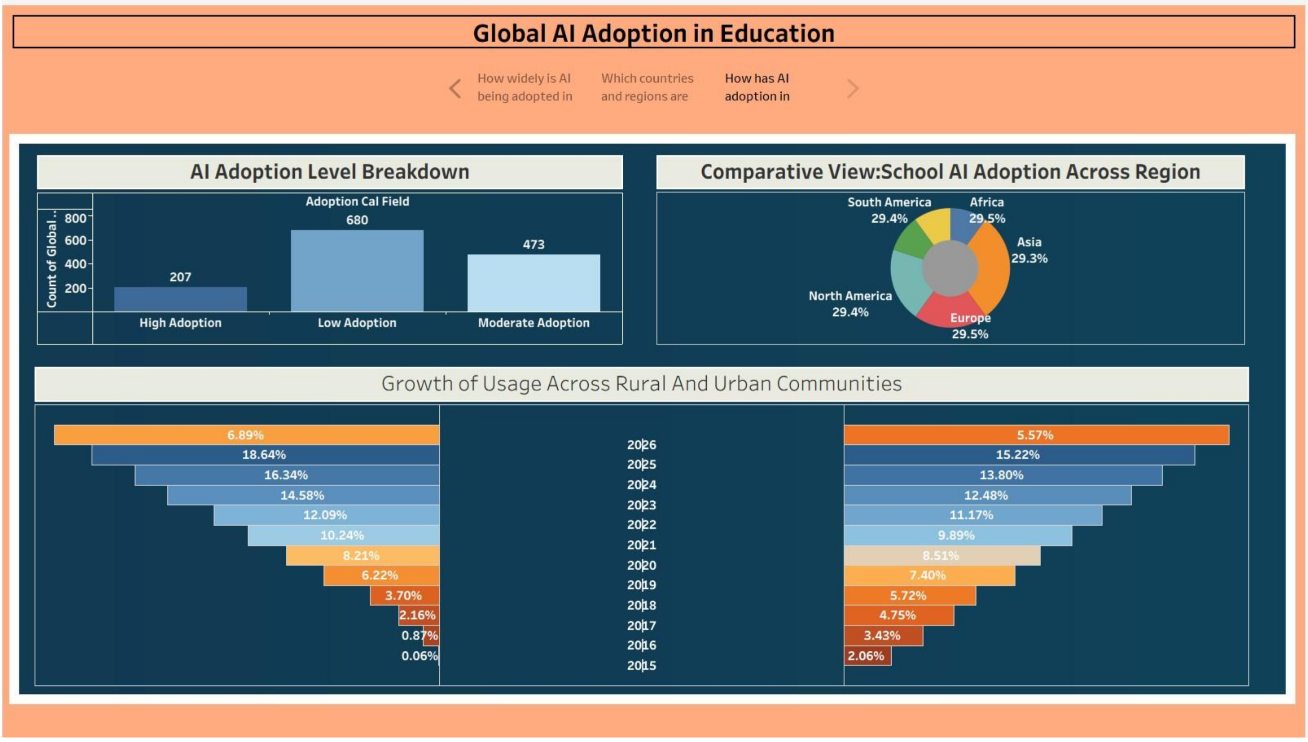
Google Gemini 24.265%
Year

3.


Global AI Adoption in Education
How widely is AI being adopted in
Which countries and regions are
How has AI adoption in
AI Adoption Level Breakdown
Comparative View:School AI Adoption Across Region
Adoption Cal Field
High Adoption
Low Adoption
Moderate Adoption
South America 29.4%
Africa 29.5%
Asia 29.3%
North America 29.4%
Europe 29.5%
Growth of Usage Across Rural And Urban Communities